

TMLR Rebuttal & Full-Scale Experiments

Three reviews received, point-by-point rebuttal written with new experimental evidence. Now revising the paper while running large-model experiments in parallel.

PRESENTING TO

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PAPER STATUS

Under Revision · TMLR

LIVE EXPERIMENT

Qwen 2.5 · 14B

3 Reviewers

Rebuttal Submitted

14B Runs In Progress

Paper Revision Ongoing

OpenReview Platform

01

How We Got Here

1

Paper submitted via OpenReview — A* search vs. CoT reasoning across 3 QA datasets (HotpotQA, StrategyQA, LogicQA). Core claim: branching complexity predicts which method wins.

2

3 reviews received — mix of borderline (2.5) and resubmit (1.5) scores. Concerns: model scale, compute cost, heuristic sensitivity, router limitations.

3

Rebuttal written — every weakness addressed with new experiments: BFS/Greedy/A* ablations, budget sweeps, token cost tables, topology analysis (26.1% oracle gap recovery).

4

Now — integrating rebuttal evidence into the revised paper. Running qwen2.5:14b full-dataset experiments (2290 + 2290 + 651 questions) across all 3 datasets.

Borderline Accept

Overall 2.5

Soundness 3

Confidence 4

- CONCERN

Routing gap between oracle and actual router is large — is this a fundamental flaw in the work?

- OUR RESPONSE

It's a **finding**, not a flaw. We quantified the Fluency-Correctness Asymmetry: 38% of divergent cases had fluent-but-wrong CoT traces — a hard ceiling for any output-based router.

- CONCERN

Heuristic may be dataset-specific and not generalise across domains.

- OUR RESPONSE

Validated on 500 annotated states across all 3 datasets. Max overestimation $\epsilon = 0.14$ was **consistent** — not dataset-dependent.

- CONCERN

No sensitivity analysis on search budget B or branching factor K.

- OUR RESPONSE

Budget sweep $B \in \{5..30\}$: peaked at $B=20$ (+5.7pp over $B=5$). Branching sweep $K \in \{1..4\}$: $K=4$ hit 80.3%, even $K=1$ gave 77.7% — robust across all settings.

Resubmit

Overall 1.5

Soundness 2.5

Confidence 4

• CONCERN

Models too small (8B / 0.5B) — conclusions may not hold at scale.

• CONCERN

Router uses handcrafted features — not portable or scalable.

• CONCERN

No compute / token cost analysis against simpler baselines.

• OUR RESPONSE

Claim is structural, not absolute. LogicQA resists search at **both** model sizes. HotpotQA gain 3.5× on smaller model confirms capability amplification — a scale-independent finding.

• OUR RESPONSE

Topology router uses 13 pre-generation features, **zero LLM calls**, and recovers 26.1% of oracle gap — beating DeepSeek-as-critic that reads full traces.

• OUR RESPONSE

A* K=1 (3,372 tok/q) already beats SC at 77.7% vs 78.2%. StrategyQA +10.3pp is **unreachable by CoT sampling alone** — SC peaks at 71.2%, still 3.6pp below A*.

Borderline · High Excitement

Overall 2.5

Excitement 3.5

Confidence 3

• CONCERN

LLM judge is DeepSeek — results may be entangled with that specific model.

• CONCERN

Only one heuristic — performance may be heuristic-dependent.

• CONCERN

Only QA benchmarks — unclear if findings generalise to math or code.

• OUR RESPONSE

BFS ablation (DeepSeek never consulted): 77.0%. Greedy: 77.7%. Full A*: 76.0%. All within **1.7pp** — the judge is not load-bearing. Branching structure drives the gain.

• OUR RESPONSE

Framework: $h=0 \rightarrow$ BFS; $g=0 \rightarrow$ Greedy; full $\rightarrow$ A*. Ablation shows all variants perform similarly — heuristic choice is **not the driver**.

• OUR RESPONSE

13 topology features are text-level and domain-agnostic. Framework predicts search helps on multi-step math — **confirmed by rStar-Math 2025**.

Who's Doing What

PAPER WRITING

Dibyanayan

- › Integrate rebuttal ablations and tables into TMLR revision.
- › Add topology-routing and compute-efficiency sections.
- › Update related-work with 5 new 2025 citations.

EXPERIMENTS

Tanzeel Khan

- › Complete CoT + A* full-dataset runs on qwen2.5:14b.
- › Produce clean comparison tables (CoT vs A*, 14B scale).
- › Prepare reasoning traces for topic modelling analysis.

3/3

Reviewers fully addressed

14B

Model running now (A100)

Rev.

TMLR paper under revision

Next

Topic modelling on traces